Cell Tuition
1-1 PRIVATE TUTORS

STATISTICS

Two-Way Tables

Name _____ Date _____ Calculator allowed Version 1



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KEY WORDS two-way table *(tūtohi ara-rua)* · row · column · total *(tapeke)* · fraction *(hautanga)*

A Completing a table ●○○

Remember: Each row adds across to its total, and each column adds down to its total.

1. Fill in the missing numbers in each table.

a)

	Walk	Bus	Car	Total
Girls	12		7	28
Boys	8	14	10	
Total		23		

b)

	Summer	Winter	Neither	Total
Year 9		27	19	
Year 10	41		17	80
Total	75	49		

B Reading a table ●○○

2. Some students were asked which pet they have at home. The table shows the results.

	Cat	Dog	No pet	Total
Girls	14	18	8	40
Boys	9	21	10	40
Total	23	39	18	80

a) How many girls have a dog? _____

b) How many boys have a cat? _____

c) How many students have no pet? _____

d) How many boys were asked? _____

e) How many students were asked altogether? _____

f) How many students have a cat or a dog? _____

C Fractions and percentages ●●○

Watch out: Read the question carefully to choose the right total: "of the students" uses the grand total, but "of the girls" uses the girls' total.

3. Use the pets table above. Give each answer as a fraction in its simplest form.

a) What fraction of the students are girls?

b) What fraction of the students have a dog?

c) What fraction of the boys have no pet?

d) What fraction of the cat owners are girls?

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4. Use the pets table above. Give each answer as a percentage, to 1 decimal place where needed.

a) What percentage of the students have no pet?

b) What percentage of the girls have a dog?

c) What percentage of the boys have a dog?

d) What percentage of the dog owners are boys?

D Probabilities from a table ●●●

Remember: Probability = the number in the group you want ÷ the total number you are choosing from.

5. The table shows whether the students in two year groups play a musical instrument. One student is chosen at random. Find the probability that the student:

	Plays an instrument	Does not play	Total
Year 9	18	42	60
Year 10	24	36	60
Total	42	78	120

a) is in Year 9

b) plays an instrument

c) is in Year 10 and plays an instrument

d) is in Year 9 and does not play an instrument

6. Use the same table.

a) A Year 9 student is chosen at random. Find the probability that this student plays an instrument.

b) A student who plays an instrument is chosen at random. Find the probability that this student is in Year 10.

7. The table shows the adults at a workplace. One adult is chosen at random. Find the probability that the adult:

	Wears glasses	No glasses	Total
Left-handed	6	14	20
Right-handed	30	70	100
Total	36	84	120

a) is left-handed

b) wears glasses

c) is right-handed and does not wear glasses

d) wears glasses, if we already know that the adult is left-handed

★ Extension

8. In the music table above, are Year 9 or Year 10 students more likely to play an instrument? Use percentages to explain.

9. There are 80 students in a group, and 45 of them are girls. 50 of the students play a sport, and 26 of these are boys. Draw a two-way table and fill in every number.

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Two-Way Tables

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

KEY WORDS two-way table *(tūtohi ara-rua)* · row · column · total *(tapeke)* · fraction *(hautanga)*

A Completing a table ●○○

Remember: Each row adds across to its total, and each column adds down to its total.

1. Fill in the missing numbers in each table.

a)

	Walk	Bus	Car	Total
Girls	11		6	30
Boys	9	12	9	
Total		25		

b)

	Summer	Winter	Neither	Total
Year 9		25	17	
Year 10	36		15	80
Total	74	54		

B Reading a table ●○○

2. Some students were asked which pet they have at home. The table shows the results.

	Cat	Dog	No pet	Total
Girls	16	15	9	40
Boys	11	20	9	40
Total	27	35	18	80

a) How many girls have a dog? _____

b) How many boys have a cat? _____

c) How many students have no pet? _____

d) How many boys were asked? _____

e) How many students were asked altogether? _____

f) How many students have a cat or a dog? _____

C Fractions and percentages ●●○

Watch out: Read the question carefully to choose the right total: "of the students" uses the grand total, but "of the girls" uses the girls' total.

3. Use the pets table above. Give each answer as a fraction in its simplest form.

a) What fraction of the students are girls?

b) What fraction of the students have a dog?

c) What fraction of the boys have no pet?

d) What fraction of the cat owners are girls?

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4. Use the pets table above. Give each answer as a percentage, to 1 decimal place where needed.

a) What percentage of the students have no pet?

b) What percentage of the girls have a dog?

c) What percentage of the boys have a dog?

d) What percentage of the dog owners are boys?

D Probabilities from a table ●●●

Remember: Probability = the number in the group you want ÷ the total number you are choosing from.

5. The table shows whether the students in two year groups play a musical instrument. One student is chosen at random. Find the probability that the student:

	Plays an instrument	Does not play	Total
Year 9	15	45	60
Year 10	27	33	60
Total	42	78	120

a) is in Year 9

b) plays an instrument

c) is in Year 10 and plays an instrument

d) is in Year 9 and does not play an instrument

6. Use the same table.

a) A Year 9 student is chosen at random. Find the probability that this student plays an instrument.

b) A student who plays an instrument is chosen at random. Find the probability that this student is in Year 10.

7. The table shows the adults at a workplace. One adult is chosen at random. Find the probability that the adult:

	Wears glasses	No glasses	Total
Left-handed	9	21	30
Right-handed	36	54	90
Total	45	75	120

a) is left-handed

b) wears glasses

c) is right-handed and does not wear glasses

d) wears glasses, if we already know that the adult is left-handed

★ Extension

8. In the music table above, are Year 9 or Year 10 students more likely to play an instrument? Use percentages to explain.

9. There are 90 students in a group, and 50 of them are girls. 55 of the students play a sport, and 24 of these are boys. Draw a two-way table and fill in every number.

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